The space L^∞

REAL VARIABLES I

Fall 2026

In Notes 24 and 25, we established the properties of the Lebesgue spaces $L^p(X,\mu)$ for finite exponents $p \in (0,+\infty)$, measuring the size of a function through the integral functional $\int_X |f|^p \, d\mu$. In Note 24, Hölder's inequality paired $L^p(X,\mu)$ with $L^{p'}(X,\mu)$ for conjugate exponents satisfying $1/p + 1/p' = 1$ with $p \in (1,+\infty)$. At the endpoint $p = 1$, the formal conjugate relation suggests $p' = +\infty$, which corresponds to bounding the pointwise magnitude of a function rather than computing an integral power. However, because elements of $L^p(X,\mu)$ are equivalence classes of functions defined up to almost-everywhere equality, the classical supremum $\sup_{x\in X} |f(x)|$ is incompatible with the measure space structure: modifying a function on a non-empty set of measure zero preserves its integrals and its L^p equivalence class, but can arbitrarily alter its supremum.

To extend the L^p framework to the endpoint exponent, we define the **essential supremum** $\operatorname{ess\,sup} |f|$, which determines the minimal bound satisfied by $|f|$ outside a set of measure zero. The resulting space $L^\infty(X,\mu)$ consists of almost-everywhere equivalence classes of essentially bounded functions. In these notes, we establish four primary results for $L^\infty(X,\mu)$. First, we prove that the essential supremum is attained as an almost-everywhere bound, and that $\|\cdot\|_\infty$ defines a norm under which $L^\infty(X,\mu)$ is a normed vector space. Second, we extend Hölder's inequality to the conjugate pair $(1,\infty)$, establishing that $L^\infty(X,\mu)$ acts as a space of bounded linear functionals on $L^1(X,\mu)$, with an isometric identification when the underlying measure is σ-finite. Third, we prove that convergence in $L^\infty(X,\mu)$ is equivalent to uniform convergence outside a single null set, and we deduce that $L^\infty(X,\mu)$ is a Banach space. Fourth, on a measure space of finite measure, we prove that $\|f\|_p \to \|f\|_\infty$ as $p \to +\infty$, providing an analytical justification for the notation L^∞.

After studying these notes, you will be able to do the following.

- Define the essential supremum $\operatorname{ess\,sup} |f|$ of a measurable function, compute it in concrete examples, and prove that $|f(x)| \le \|f\|_\infty$ almost everywhere whenever $\|f\|_\infty < +\infty$.
- Construct the space $L^\infty(X,\mu)$ of almost-everywhere equivalence classes of essentially bounded functions, and verify that $\|\cdot\|_\infty$ satisfies the separation, absolute homogeneity, and triangle inequality axioms of a norm.
- State and prove Hölder's inequality for the conjugate exponent pair $(1,\infty)$, and determine the operator norm of the induced functional $\Lambda_g \in (L^1)^*$ on σ-finite measure spaces.
- Characterize metric convergence in $L^\infty(X,\mu)$ as uniform convergence outside a single null set, and prove that the normed space $(L^\infty(X,\mu), \|\cdot\|_\infty)$ is complete.
- Prove that on a finite measure space, $L^\infty(X,\mu) \subseteq L^p(X,\mu)$ for all $p \in (0,+\infty)$, and prove the limit formula $\lim_{p\to+\infty} \|f\|_p = \|f\|_\infty$.

Throughout these notes, $(X,\mathcal{A},\mu)$ denotes a measure space and functions are measurable functions $f\colon X \to \mathbb{R}$. On subsets of $\mathbb{R}$, the symbol $\mathcal{L}$ denotes Lebesgue measure.

13.1 The essential supremum

For a measurable function $f\colon X \to \mathbb{R}$, a real number $M \geq 0$ is an **almost-everywhere bound** for $|f|$ if $|f(x)| \leq M$ for almost every point $x \in X$, or equivalently, if the superlevel set

$$\{x \in X : |f(x)| > M\}$$

satisfies $\mu(\{x \in X : |f(x)| > M\}) = 0$. The essential supremum of $|f|$ is the infimum of all almost-everywhere bounds.

Definition 13.1 (Essential supremum). The **essential supremum** of a measurable function $f\colon X \to \mathbb{R}$ is

$$\|f\|_\infty \coloneqq \operatorname*{ess\,sup}_{x \in X} |f(x)| \coloneqq \inf\{M \in [0, +\infty) : \mu(\{x \in X : |f(x)| > M\}) = 0\},$$

under the convention that $\inf \emptyset = +\infty$. The function f is **essentially bounded** if $\|f\|_\infty < +\infty$.

The collection of almost-everywhere bounds has a hereditary structure: if M is an almost-everywhere bound and $M' \geq M$, then $\{x \in X : |f(x)| > M'\} \subseteq \{x \in X : |f(x)| > M\}$. By monotonicity of the measure μ,

$$\mu(\{|f| > M'\}) \leq \mu(\{|f| > M\}) = 0,$$

so M' is also an almost-everywhere bound. Consequently, the set of almost-everywhere bounds is an interval extending to $+\infty$. The next lemma proves that this interval contains its infimum, so that the infimum in Definition 13.1 is attained as a minimum.

Lemma 13.2 (The essential supremum is an almost-everywhere bound). Suppose that $f\colon X \to \mathbb{R}$ is a measurable function with $\|f\|_\infty < +\infty$. Then $|f(x)| \leq \|f\|_\infty$ for almost every $x \in X$.

Proof of Lemma 13.2. We claim that the set $\{x \in X : |f(x)| > \|f\|_\infty\}$ has measure zero. For each positive integer $k \in \mathbb{N}$, the number $\|f\|_\infty + 1/k$ strictly exceeds the infimum in Definition 13.1. By definition of the infimum, there exists an almost-everywhere bound $M_k \in [0, +\infty)$ such that

$$\|f\|_\infty \leq M_k < \|f\|_\infty + \frac{1}{k}.$$

Because M_k is an almost-everywhere bound, we have $\mu(\{|f| > M_k\}) = 0$. Since $\{|f| > \|f\|_\infty + 1/k\} \subseteq \{|f| > M_k\}$, monotonicity of μ implies that

$$\mu\left(\left\{x \in X : |f(x)| > \|f\|_\infty + \frac{1}{k}\right\}\right) = 0 \qquad \text{for every } k \in \mathbb{N}.$$

A real number strictly exceeds $\|f\|_\infty$ if and only if it strictly exceeds $\|f\|_\infty + 1/k$ for some positive integer $k \in \mathbb{N}$. Therefore, the superlevel set decomposes as a countable union:

$$\{x \in X : |f(x)| > \|f\|_\infty\} = \bigcup_{k=1}^{+\infty} \left\{x \in X : |f(x)| > \|f\|_\infty + \frac{1}{k}\right\}.$$

Applying countable subadditivity of the measure μ yields

$$\mu(\{x \in X : |f(x)| > \|f\|_\infty\}) \leq \sum_{k=1}^{+\infty} \mu\left(\left\{x \in X : |f(x)| > \|f\|_\infty + \frac{1}{k}\right\}\right) = \sum_{k=1}^{+\infty} 0 = 0.$$

This proves that $|f(x)| \leq \|f\|_\infty$ for almost every $x \in X$. □

Because the pointwise supremum $\sup_{x \in X} |f(x)|$ bounds $|f(x)|$ at every point of X, it is an almost-everywhere bound, and therefore

$$\|f\|_\infty \leq \sup_{x \in X} |f(x)|.$$

The essential supremum can be strictly smaller than the pointwise supremum, as illustrated in the following examples.

Example 13.3 (Comparison between essential supremum and supremum). We examine the essential supremum in four standard settings.

Modification on a null set. On $\mathbb{R}$ equipped with Lebesgue measure, consider the indicator function of the rationals, $f \coloneqq \mathbb{1}_{\mathbb{Q}}$. The pointwise supremum satisfies $\sup_{\mathbb{R}} \mathbb{1}_{\mathbb{Q}} = 1$, because $\mathbb{1}_{\mathbb{Q}}(x) = 1$ at every rational number. However, the set $\{x \in \mathbb{R} : \mathbb{1}_{\mathbb{Q}}(x) > 0\} = \mathbb{Q}$ is countable and has Lebesgue measure zero: $\mathcal{L}(\mathbb{Q}) = 0$. Therefore 0 is an almost-everywhere bound for $\mathbb{1}_{\mathbb{Q}}$, which implies that $\|\mathbb{1}_{\mathbb{Q}}\|_\infty = 0$. More generally, modifying a function on an arbitrary set of measure zero leaves its essential supremum unchanged.

Continuous functions on a compact interval. On the compact interval $[0,1]$ equipped with Lebesgue measure, suppose that $f\colon [0,1] \to \mathbb{R}$ is continuous. By the extreme value theorem, $|f|$ attains a maximum value $M_0 \coloneqq \max_{[0,1]} |f|$. We claim that $\|f\|_\infty = M_0$. Indeed, $\|f\|_\infty \leq M_0$ holds because M_0 is an everywhere bound. Conversely, for any number $M < M_0$, the set $U \coloneqq \{x \in [0,1] : |f(x)| > M\}$ is non-empty and relatively open in $[0,1]$ by continuity of $|f|$. Every non-empty relatively open subset of $[0,1]$ contains an open interval of positive length, so $\mathcal{L}(U) > 0$. Thus no number strictly smaller than M_0 can be an almost-everywhere bound, proving that $\|f\|_\infty = \max_{[0,1]} |f|$.

An integrable but essentially unbounded function. On the half-open interval $(0,1]$ equipped with Lebesgue measure, consider the power function $f(x) \coloneqq x^{-1/2}$. The function f is integrable: $\int_{(0,1]} x^{-1/2} \, \mathrm{d}\mathcal{L} = \lim_{\varepsilon \to 0^+} [2x^{1/2}]_\varepsilon^1 = 2 < +\infty$, so $f \in L^1((0,1], \mathcal{L})$. However, for every threshold $M > 0$, the set

$$\{x \in (0,1] : x^{-1/2} > M\} = (0, \min\{1, M^{-2}\})$$

has strictly positive Lebesgue measure $\min\{1, M^{-2}\} > 0$. Therefore no finite number M is an almost-everywhere bound, and $\|x^{-1/2}\|_\infty = +\infty$.

Counting measure and sequence spaces. On the set of positive integers $\mathbb{N}$ equipped with counting measure μ_{count}, the only set of measure zero is the empty set: $\mu_{\text{count}}(E) = 0$ if and only if $E = \emptyset$. Consequently, a bound holds almost everywhere with respect to counting measure if and only if it holds at every point of $\mathbb{N}$. For every sequence $a = (a_k)_{k \in \mathbb{N}}$, the essential supremum coincides with the supremum:

$$\|a\|_\infty = \sup_{k \in \mathbb{N}} |a_k|.$$

13.2 The normed space L^∞

To define a normed space whose elements can be distinguished by the functional $\|\cdot\|_\infty$, we identify functions that agree almost everywhere, following the construction of $L^p(X, \mu)$ in Note 24. Suppose that $f, g\colon X \to \mathbb{R}$ are measurable functions with $f = g$ almost everywhere on X, and define the null set $N \coloneqq \{x \in X : f(x) \neq g(x)\}$. For every $M \geq 0$, the symmetric difference of the superlevel sets satisfies

$$\{x \in X : |f(x)| > M\} \triangle \{x \in X : |g(x)| > M\} \subseteq N.$$

Because $\mu(N) = 0$, the two superlevel sets have identical measure:

$$\mu(\{|f| > M\}) = \mu(\{|g| > M\}).$$

Consequently, M is an almost-everywhere bound for $|f|$ if and only if it is an almost-everywhere bound for $|g|$. Taking the infimum over all such bounds yields $\|f\|_\infty = \|g\|_\infty$. The essential supremum is therefore independent of the choice of representative within an almost-everywhere equivalence class.

Definition 13.4 (The space L^∞). We define the vector space of essentially bounded measurable functions by

$$\mathcal{L}^\infty(X,\mu) \coloneqq \{f\colon X \to \mathbb{R} : f \text{ is measurable and } \|f\|_\infty < +\infty\}.$$

The space $L^\infty(X,\mu)$ is the quotient space of $\mathcal{L}^\infty(X,\mu)$ with respect to the equivalence relation of almost-everywhere equality:

$$L^\infty(X,\mu) \coloneqq \mathcal{L}^\infty(X,\mu) \,/\, \sim .$$

For an equivalence class $[f] \in L^\infty(X,\mu)$, the vector space operations and the functional $\|[f]\|_\infty \coloneqq \|f\|_\infty$ are defined through any representative function $f \in [f]$. As in Note 24, we write f rather than $[f]$ when no ambiguity arises. When $X = \mathbb{N}$ equipped with counting measure, this space is denoted ℓ^∞; it is the classical space of bounded real sequences.

Proposition 13.5 (L^∞ is a normed space). The space $L^\infty(X,\mu)$ is a real vector space, and $\|\cdot\|_\infty$ is a norm on it: for all $f, g \in L^\infty(X,\mu)$ and $\lambda \in \mathbb{R}$,

$$\|f\|_\infty = 0 \iff f = 0 \quad \text{in } L^\infty(X,\mu), \tag{13.1}$$

$$\|\lambda f\|_\infty = |\lambda|\,\|f\|_\infty, \tag{13.2}$$

$$\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty. \tag{13.3}$$

Proof of Proposition 13.5. Step 1: Separation axiom.
Suppose that $\|f\|_\infty = 0$. By Lemma 13.2, $|f(x)| \le \|f\|_\infty = 0$ for almost every $x \in X$. Because $|f(x)| \ge 0$ everywhere, it follows that $f(x) = 0$ almost everywhere on X, which means that $f = 0$ in $L^\infty(X,\mu)$. Conversely, if $f = 0$ in $L^\infty(X,\mu)$, then $f = 0$ almost everywhere. Then 0 is an almost-everywhere bound for $|f|$, which implies that $\|f\|_\infty \le 0$. Since $\|f\|_\infty \ge 0$ by definition, we conclude that $\|f\|_\infty = 0$.

Step 2: Absolute homogeneity.
For the scalar $\lambda = 0$, the function $0 \cdot f$ is the zero function, so both sides of (13.2) vanish. Suppose that $\lambda \neq 0$. For every number $M \ge 0$, the pointwise identity $|\lambda f(x)| = |\lambda|\,|f(x)|$ gives

$$\{x \in X : |\lambda f(x)| > |\lambda| M\} = \{x \in X : |f(x)| > M\}.$$

Therefore, M is an almost-everywhere bound for $|f|$ if and only if $|\lambda| M$ is an almost-everywhere bound for $|\lambda f|$. Taking the infimum over all almost-everywhere bounds M yields $\|\lambda f\|_\infty = |\lambda|\,\|f\|_\infty$.

Step 3: Triangle inequality and vector space closure.
Suppose that $f, g \in L^\infty(X,\mu)$. By Lemma 13.2, there exist measurable sets $N_f, N_g \subseteq X$ with $\mu(N_f) = \mu(N_g) = 0$ such that $|f(x)| \le \|f\|_\infty$ for all $x \in X \setminus N_f$ and $|g(x)| \le \|g\|_\infty$ for all $x \in X \setminus N_g$. We define the union $N \coloneqq N_f \cup N_g$. By subadditivity of the measure, $\mu(N) \le \mu(N_f) + \mu(N_g) = 0$. For every point $x \in X \setminus N$, the triangle inequality in $\mathbb{R}$ yields

$$|f(x) + g(x)| \le |f(x)| + |g(x)| \le \|f\|_\infty + \|g\|_\infty.$$

Because $\mu(N) = 0$, the finite real number $\|f\|_\infty + \|g\|_\infty$ is an almost-everywhere bound for $|f+g|$. By Definition 13.1, it follows that

$$\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty < +\infty.$$

Hence $f + g \in \mathcal{L}^\infty(X,\mu)$, and the triangle inequality (13.3) holds. Together with Step 2, this establishes that $\mathcal{L}^\infty(X,\mu)$ is a real vector space, and that $\|\cdot\|_\infty$ defines a norm on the quotient space $L^\infty(X,\mu)$. □

13.2.1 Hölder's inequality at the endpoint

In Note 24, conjugate exponents $p, p' \in (1, +\infty)$ satisfied the reciprocal relation $1/p + 1/p' = 1$. Extending this relation to the endpoint $p = 1$ under the convention $1/\infty \coloneqq 0$ pairs the exponent 1 with ∞: the conjugate exponent of 1 is ∞, and the conjugate exponent of ∞ is 1. With this convention, Hölder's inequality takes a direct form.

Proposition 13.6 (Hölder's inequality for the exponents 1 and ∞). Suppose that $f \in L^1(X, \mu)$ and $g \in L^\infty(X, \mu)$. Then the product function fg belongs to $L^1(X, \mu)$ and

$$\int_X |fg| \, \mathrm{d}\mu \leq \|f\|_1 \, \|g\|_\infty. \tag{13.4}$$

Proof of Proposition 13.6. By Lemma 13.2, there exists a measurable set $N \subseteq X$ with $\mu(N) = 0$ such that $|g(x)| \leq \|g\|_\infty$ for all $x \in X \setminus N$. Multiplying both sides by the non-negative quantity $|f(x)|$ yields

$$|f(x)g(x)| \leq \|g\|_\infty |f(x)| \qquad \text{for all } x \in X \setminus N.$$

Because this inequality holds almost everywhere on X, monotonicity and positive linearity of the Lebesgue integral (Note 19 / Note 21) yield

$$\int_X |fg| \, \mathrm{d}\mu \leq \int_X \|g\|_\infty |f| \, \mathrm{d}\mu = \|g\|_\infty \int_X |f| \, \mathrm{d}\mu = \|f\|_1 \, \|g\|_\infty.$$

Because $\|f\|_1 < +\infty$ and $\|g\|_\infty < +\infty$, the integral $\int_X |fg| \, \mathrm{d}\mu$ is finite, so $fg \in L^1(X, \mu)$. □

By (13.4), each element $g \in L^\infty(X, \mu)$ induces a linear functional $\Lambda_g \colon L^1(X, \mu) \to \mathbb{R}$ defined by

$$\Lambda_g(f) \coloneqq \int_X fg \, \mathrm{d}\mu.$$

Hölder's inequality implies that $|\Lambda_g(f)| \leq \int_X |fg| \, \mathrm{d}\mu \leq \|g\|_\infty \|f\|_1$, so Λ_g is a bounded linear functional on the normed space $(L^1(X, \mu), \|\cdot\|_1)$ with operator norm

$$\|\Lambda_g\|_{(L^1)^*} \coloneqq \sup_{\substack{f \in L^1(X,\mu) \\ \|f\|_1 \leq 1}} |\Lambda_g(f)| \leq \|g\|_\infty.$$

When the underlying measure space is σ-finite, this bound is an equality, which establishes that the map $g \mapsto \Lambda_g$ is an isometric embedding of $L^\infty(X, \mu)$ into the dual space $(L^1(X, \mu))^*$.

Proposition 13.7 (Duality isometry for L^∞ on σ-finite measure spaces). Suppose that $(X, \mathcal{A}, \mu)$ is a σ-finite measure space. For every $g \in L^\infty(X, \mu)$, the induced linear functional $\Lambda_g \in (L^1(X, \mu))^*$ satisfies

$$\|\Lambda_g\|_{(L^1)^*} = \|g\|_\infty. \tag{13.5}$$

Proof of Proposition 13.7. Step 1: The upper bound.
For every $f \in L^1(X, \mu)$ with $\|f\|_1 \leq 1$, Proposition 13.6 gives

$$|\Lambda_g(f)| = \left| \int_X fg \, \mathrm{d}\mu \right| \leq \int_X |fg| \, \mathrm{d}\mu \leq \|f\|_1 \, \|g\|_\infty \leq \|g\|_\infty.$$

Taking the supremum over all such functions f yields $\|\Lambda_g\|_{(L^1)^*} \leq \|g\|_\infty$.

Step 2: The lower bound.
If $\|g\|_\infty = 0$, then $\|\Lambda_g\|_{(L^1)^*} = 0$ by Step 1. Suppose that $\|g\|_\infty > 0$. Consider an arbitrary real number ε satisfying $0 < \varepsilon < \|g\|_\infty$. By definition of the essential supremum, the set

$$A \coloneqq \{x \in X : |g(x)| > \|g\|_\infty - \varepsilon\}$$

satisfies $\mu(A) > 0$. Because $(X, \mathcal{A}, \mu)$ is σ-finite, there exists a countable sequence of measurable sets X_k of finite measure covering X. The sets $A \cap X_k$ increase to A, so continuity of measure from below implies that $\mu(A \cap X_k) > 0$ for some index k. Setting $B \coloneqq A \cap X_k$ provides a measurable subset $B \subseteq A$ satisfying $0 < \mu(B) < +\infty$. We define the test function $f_0 \colon X \to \mathbb{R}$ by

$$f_0(x) \coloneqq \frac{\operatorname{sgn}(g(x))\, \mathbb{1}_B(x)}{\mu(B)}, \qquad x \in X,$$

where $\operatorname{sgn}(t) \coloneqq 1$ for $t \geq 0$ and $\operatorname{sgn}(t) \coloneqq -1$ for $t < 0$. The function f_0 belongs to $L^1(X, \mu)$ and satisfies

$$\|f_0\|_1 = \int_X \frac{|\operatorname{sgn}(g)|\, \mathbb{1}_B}{\mu(B)}\, \mathrm{d}\mu = \frac{1}{\mu(B)} \int_B 1\, \mathrm{d}\mu = 1.$$

Evaluating the functional on f_0 yields

$$\Lambda_g(f_0) = \int_X g f_0\, \mathrm{d}\mu = \frac{1}{\mu(B)} \int_B g(x) \operatorname{sgn}(g(x))\, \mathrm{d}\mu = \frac{1}{\mu(B)} \int_B |g|\, \mathrm{d}\mu.$$

Because $|g(x)| > \|g\|_\infty - \varepsilon$ for every point $x \in B$, monotonicity of the integral gives

$$\Lambda_g(f_0) \geq \frac{1}{\mu(B)} \int_B (\|g\|_\infty - \varepsilon)\, \mathrm{d}\mu = \|g\|_\infty - \varepsilon.$$

Because $\|f_0\|_1 = 1$, it follows that $\|\Lambda_g\|_{(L^1)^*} \geq \Lambda_g(f_0) \geq \|g\|_\infty - \varepsilon$. Because $\varepsilon > 0$ was arbitrary, we conclude that $\|\Lambda_g\|_{(L^1)^*} \geq \|g\|_\infty$. Together with Step 1, this establishes (13.5). □

13.3 Convergence and completeness

A sequence $(f_n)_{n \in \mathbb{N}}$ in $L^\infty(X, \mu)$ **converges in** L^∞ to $f \in L^\infty(X, \mu)$ if

$$\lim_{n \to \infty} \|f_n - f\|_\infty = 0.$$

For bounded functions on an arbitrary set, the supremum metric $\sup_{x \in X} |f_n(x) - f(x)|$ describes uniform convergence. The essential supremum describes the same uniform behavior after discarding a single set of measure zero.

Proposition 13.8 (L^∞ convergence is uniform convergence outside a null set). For a sequence $(f_n)_{n \in \mathbb{N}}$ in $L^\infty(X, \mu)$ and an element $f \in L^\infty(X, \mu)$, the following conditions are equivalent:

1. $\lim_{n \to \infty} \|f_n - f\|_\infty = 0$.
2. There exists a measurable set $N \subseteq X$ with $\mu(N) = 0$ such that $f_n \to f$ uniformly on $X \setminus N$ as $n \to \infty$.

Proof of Proposition 13.8. Step 1: Convergence in L^∞ implies uniform convergence outside a

null set.
Suppose that $\|f_n - f\|_\infty \to 0$ as $n \to \infty$. For each index $n \in \mathbb{N}$, Lemma 13.2 implies that the set

$$N_n \coloneqq \{x \in X : |f_n(x) - f(x)| > \|f_n - f\|_\infty\}$$

satisfies $\mu(N_n) = 0$. We define the countable union $N \coloneqq \bigcup_{n=1}^{+\infty} N_n$. By countable subadditivity of the measure μ,

$$\mu(N) \leq \sum_{n=1}^{+\infty} \mu(N_n) = \sum_{n=1}^{+\infty} 0 = 0.$$

For every point $x \in X \setminus N$ and every index $n \in \mathbb{N}$, we have $x \notin N_n$, which implies that

$$|f_n(x) - f(x)| \leq \|f_n - f\|_\infty.$$

Taking the supremum over all points $x \in X \setminus N$ yields

$$\sup_{x \in X \setminus N} |f_n(x) - f(x)| \leq \|f_n - f\|_\infty \longrightarrow 0 \qquad \text{as } n \to \infty.$$

This proves that $f_n \to f$ uniformly on $X \setminus N$.

Step 2: Uniform convergence outside a null set implies convergence in L^∞.
Suppose that there exists a measurable set $N \subseteq X$ with $\mu(N) = 0$ such that $f_n \to f$ uniformly on $X \setminus N$. For each $n \in \mathbb{N}$, the quantity $S_n \coloneqq \sup_{x \in X \setminus N} |f_n(x) - f(x)|$ is a finite real number, and $|f_n(x) - f(x)| \leq S_n$ for all $x \in X \setminus N$. Because $\mu(N) = 0$, the number S_n is an almost-everywhere bound for $|f_n - f|$. By Definition 13.1, it follows that

$$\|f_n - f\|_\infty \leq S_n = \sup_{x \in X \setminus N} |f_n(x) - f(x)|.$$

Because $f_n \to f$ uniformly on $X \setminus N$, the supremum S_n tends to 0 as $n \to \infty$. Therefore $\|f_n - f\|_\infty \to 0$ as $n \to \infty$. □

A notable distinction between L^∞ convergence and L^p convergence for finite exponents $p < +\infty$ concerns pointwise convergence. In Note 25, convergence in L^p guaranteed pointwise convergence only along a subsequence (Corollary 25.4 / Note 25), and the full typewriter sequence failed to converge at every point of the domain (Example 25.5 / Note 25). In contrast, Proposition 13.8 shows that convergence in $L^\infty(X, \mu)$ implies that the full sequence $(f_n)_{n \in \mathbb{N}}$ converges pointwise outside the single null set N.

Conversely, almost-everywhere convergence does not imply L^∞ convergence. For example, on $\mathbb{R}$ equipped with Lebesgue measure, consider the sequence of translated indicator functions $f_n \coloneqq \mathbb{1}_{[n, n+1)}$ for $n \in \mathbb{N}$. For every fixed $x \in \mathbb{R}$, we have $x < n$ for all sufficiently large n, so $f_n(x) = 0$. Thus $f_n(x) \to 0$ for every $x \in \mathbb{R}$. However, the essential supremum remains constant:

$$\|f_n - 0\|_\infty = \|\mathbb{1}_{[n, n+1)}\|_\infty = 1 \qquad \text{for all } n \in \mathbb{N},$$

so (f_n) does not converge in $L^\infty(\mathbb{R}, \mathcal{L})$.

Theorem 13.9 (Completeness of L^∞). The normed space $(L^\infty(X, \mu), \|\cdot\|_\infty)$ is complete: every Cauchy sequence in $L^\infty(X, \mu)$ converges in L^∞ to an element of $L^\infty(X, \mu)$. Consequently, $L^\infty(X, \mu)$ is a Banach space.

Proof of Theorem 13.9. Suppose that $(f_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $L^\infty(X, \mu)$. We choose a measurable representative function $f_n : X \to \mathbb{R}$ for each element of the sequence.

Step 1: Construction of a common null set for the Cauchy property.
For each pair of indices $n, m \in \mathbb{N}$, Lemma 13.2 implies that the set

$$N_{n,m} \coloneqq \{x \in X : |f_n(x) - f_m(x)| > \|f_n - f_m\|_\infty\}$$

satisfies $\mu(N_{n,m}) = 0$. We define the countable union

$$N \coloneqq \bigcup_{n=1}^{+\infty} \bigcup_{m=1}^{+\infty} N_{n,m}.$$

Countable subadditivity of the measure μ implies that $\mu(N) = 0$. For every point $x \in X \setminus N$ and all indices $n, m \in \mathbb{N}$, we have $x \notin N_{n,m}$, which implies that

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty \qquad \text{for all } x \in X \setminus N. \tag{13.6}$$

Step 2: Definition of the pointwise candidate limit.
By (13.6), for every point $x \in X \setminus N$, the sequence of real numbers $(f_n(x))_{n\in\mathbb{N}}$ satisfies

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty \longrightarrow 0 \qquad \text{as } n, m \to \infty.$$

Because $\mathbb{R}$ is complete, the sequence $(f_n(x))_{n\in\mathbb{N}}$ converges to a real number as $n \to \infty$. We define the candidate limit function $f\colon X \to \mathbb{R}$ by

$$f(x) \coloneqq \begin{cases} \lim_{n\to\infty} f_n(x), & x \in X \setminus N, \\ 0, & x \in N. \end{cases}$$

The function f is measurable because it is the pointwise limit of the sequence of measurable functions $f_n \mathbb{1}_{X\setminus N}$.

Step 3: Uniform convergence outside the null set and completeness.
Suppose that $\varepsilon > 0$. Because $(f_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $L^\infty(X, \mu)$, there exists an index $K \in \mathbb{N}$ such that

$$\|f_n - f_m\|_\infty < \varepsilon \qquad \text{for all } n, m \geq K.$$

By (13.6), for every $x \in X \setminus N$ and all $n, m \geq K$,

$$|f_n(x) - f_m(x)| < \varepsilon.$$

Taking the limit as $m \to \infty$ in this numerical inequality yields

$$|f_n(x) - f(x)| \leq \varepsilon \qquad \text{for all } x \in X \setminus N \text{ and all } n \geq K.$$

Taking the supremum over all $x \in X \setminus N$ gives

$$\sup_{x\in X\setminus N} |f_n(x) - f(x)| \leq \varepsilon \qquad \text{for all } n \geq K.$$

Because $\mu(N) = 0$, the number ε is an almost-everywhere bound for $|f_n - f|$ whenever $n \geq K$. Therefore,

$$\|f_n - f\|_\infty \leq \varepsilon \qquad \text{for all } n \geq K.$$

In particular, choosing $n = K$ implies that $f - f_K \in \mathcal{L}^\infty(X, \mu)$. Because $\mathcal{L}^\infty(X, \mu)$ is a vector space (Proposition 13.5), the sum $f = f_K + (f - f_K)$ belongs to $\mathcal{L}^\infty(X, \mu)$, so $f \in L^\infty(X, \mu)$. Because $\varepsilon > 0$ was arbitrary, it follows that $\|f_n - f\|_\infty \to 0$ as $n \to \infty$. This proves that every Cauchy sequence in $L^\infty(X, \mu)$ converges in $L^\infty(X, \mu)$. □

13.4 L^∞ as the limit of L^p

On a finite measure space, the inclusion theorem of Note 24 established that $L^q(X,\mu) \subseteq L^p(X,\mu)$ whenever $0 < p < q < +\infty$. The space $L^\infty(X,\mu)$ is contained in every one of these spaces, and its norm is the limit of the L^p norms as $p \to +\infty$.

Proposition 13.10 (The L^p norms converge to the essential supremum). Suppose that $(X, \mathcal{A}, \mu)$ is a measure space with $\mu(X) < +\infty$. For every measurable function $f\colon X \to \mathbb{R}$ and every exponent $p \in (0, +\infty)$, the quantity $\|f\|_p \coloneqq \left(\int_X |f|^p \, \mathrm{d}\mu\right)^{1/p} \in [0, +\infty]$ satisfies

$$\|f\|_p \le \mu(X)^{1/p} \, \|f\|_\infty. \tag{13.7}$$

Consequently, $L^\infty(X,\mu) \subseteq L^p(X,\mu)$ for all $p \in (0, +\infty)$. Moreover,

$$\lim_{p \to +\infty} \|f\|_p = \|f\|_\infty \qquad \text{in } [0, +\infty]. \tag{13.8}$$

Proof of Proposition 13.10. If $\mu(X) = 0$, then every measurable function satisfies $f = 0$ almost everywhere, so $\|f\|_p = 0$ for all $p \in (0, +\infty)$ and $\|f\|_\infty = 0$, and both (13.7) and (13.8) hold. Suppose that $\mu(X) > 0$.

Step 1: The upper bound $\limsup_{p \to +\infty} \|f\|_p \le \|f\|_\infty$.
If $\|f\|_\infty = +\infty$, then the right-hand side of (13.7) equals $+\infty$, so the inequality holds automatically. Suppose that $\|f\|_\infty < +\infty$. By Lemma 13.2, $|f(x)| \le \|f\|_\infty$ for almost every $x \in X$. Raising both sides to the p-th power and integrating over X with respect to μ yields

$$\int_X |f|^p \, \mathrm{d}\mu \le \int_X \|f\|_\infty^p \, \mathrm{d}\mu = \|f\|_\infty^p \, \mu(X).$$

Taking the p-th root on both sides yields (13.7):

$$\|f\|_p \le \mu(X)^{1/p} \, \|f\|_\infty.$$

Because $\mu(X) \in (0, +\infty)$, we have $\lim_{p \to +\infty} \mu(X)^{1/p} = 1$. Taking the limit superior as $p \to +\infty$ on both sides of (13.7) yields

$$\limsup_{p \to +\infty} \|f\|_p \le \lim_{p \to +\infty} \mu(X)^{1/p} \, \|f\|_\infty = \|f\|_\infty. \tag{13.9}$$

This inequality holds whether $\|f\|_\infty$ is finite or infinite.

Step 2: The lower bound $\liminf_{p \to +\infty} \|f\|_p \ge \|f\|_\infty$.
If $\|f\|_\infty = 0$, then $\liminf_{p \to +\infty} \|f\|_p \ge 0 = \|f\|_\infty$ holds trivially. Suppose that $\|f\|_\infty > 0$. Consider an arbitrary real number M satisfying $0 < M < \|f\|_\infty$. Because M is strictly smaller than the essential supremum, it is not an almost-everywhere bound for $|f|$. Therefore, the superlevel set

$$A \coloneqq \{x \in X : |f(x)| > M\}$$

satisfies $0 < \mu(A) \le \mu(X) < +\infty$. Because $|f(x)| \ge M \mathbb{1}_A(x)$ at every point $x \in X$, monotonicity of the integral yields

$$\int_X |f|^p \, \mathrm{d}\mu \ge \int_X M^p \mathbb{1}_A \, \mathrm{d}\mu = M^p \mu(A).$$

Taking the p-th root on both sides gives

$$\|f\|_p \ge M \, \mu(A)^{1/p}.$$

Because $\mu(A) \in (0, +\infty)$, we have $\lim_{p \to +\infty} \mu(A)^{1/p} = 1$. Taking the limit inferior as $p \to +\infty$ yields

$$\liminf_{p \to +\infty} \|f\|_p \geq \lim_{p \to +\infty} M \mu(A)^{1/p} = M.$$

Because M was an arbitrary number strictly smaller than $\|f\|_\infty$, we may take the supremum over all such $M < \|f\|_\infty$ to obtain

$$\liminf_{p \to +\infty} \|f\|_p \geq \|f\|_\infty. \tag{13.10}$$

This bound holds both when $\|f\|_\infty < +\infty$ and when $\|f\|_\infty = +\infty$. Combining (13.9) and (13.10) establishes that $\lim_{p \to +\infty} \|f\|_p = \|f\|_\infty$. □

The assumption that $\mu(X) < +\infty$ is essential for the inclusion $L^\infty(X, \mu) \subseteq L^p(X, \mu)$. On a measure space of infinite measure, an essentially bounded function can fail to belong to $L^p(X, \mu)$ for any finite exponent. For example, on $\mathbb{R}$ equipped with Lebesgue measure, the constant function $f(x) \equiv 1$ belongs to $L^\infty(\mathbb{R}, \mathcal{L})$ with $\|f\|_\infty = 1$, but fails to belong to $L^p(\mathbb{R}, \mathcal{L})$ for every $p \in (0, +\infty)$ because $\int_{\mathbb{R}} 1^p \, \mathrm{d}\mathcal{L} = \mathcal{L}(\mathbb{R}) = +\infty$.

Remark (Summary of properties of the space L^∞). For an arbitrary measure space $(X, \mathcal{A}, \mu)$:

- **Attained bound:** The essential supremum $\|f\|_\infty$ is the minimal almost-everywhere bound for $|f|$; that is, $|f(x)| \leq \|f\|_\infty$ for almost every $x \in X$ whenever $\|f\|_\infty < +\infty$ (Lemma 13.2).
- **Banach space:** The quotient space $L^\infty(X, \mu) = \mathcal{L}^\infty(X, \mu)/\sim$ equipped with $\|\cdot\|_\infty$ is a complete normed vector space (Proposition 13.5, Theorem 13.9).
- **Endpoint duality:** For $f \in L^1(X, \mu)$ and $g \in L^\infty(X, \mu)$, Hölder's inequality states that $\int_X |fg| \, \mathrm{d}\mu \leq \|f\|_1 \|g\|_\infty$ (Proposition 13.6); when μ is σ-finite, the map $g \mapsto \Lambda_g$ is an isometry of $L^\infty(X, \mu)$ into $(L^1(X, \mu))^*$ (Proposition 13.7).
- **Uniform convergence off a null set:** Metric convergence $\|f_n - f\|_\infty \to 0$ is equivalent to uniform convergence $f_n \to f$ outside a single set of measure zero (Proposition 13.8).
- **Asymptotic limit of L^p norms:** If $\mu(X) < +\infty$, then $L^\infty(X, \mu) \subseteq L^p(X, \mu)$ for all $p \in (0, +\infty)$, and $\lim_{p \to +\infty} \|f\|_p = \|f\|_\infty$ (Proposition 13.10).